

Umama Junejo

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EDUCATION

University of Washington

June, 2028

B.S., Industrial and Systems Engineering

Seattle, WA

Honors Interdisciplinary Student - coursework bridging science, humanities, and social sciences.

Awards: Dean's List

SKILLS

Programming: Python, Java, Arduino, Embedded Controls

Design: AutoCAD, SolidWorks, Technical drawings

Fabrication: Mill, lathe, band saw, drill press, 3D printing, laser cutting

EXPERIENCE

Undergraduate Researcher

Sep. 2025 - Present

AI Institute in Dynamic Systems Lab

- Conducted literature review on sustainable aviation fuel chemistry and machine learning to independently build the technical foundation needed to contribute to ongoing research.
- Investigated data types and simulation inputs most effective for modeling variation in aviation fuel behavior across experimental conditions.
- Developing a surrogate model using Gaussian process regression in Python to predict derived cetane number (DCN) from experimental datasets, replacing costly ignition delay testing and accelerating carbon-neutral fuel development.

Autonomous Insect Robotics Lab

- Design and prototype a servo-actuated, magnet-assisted release mechanism for dragonfly flight research using iterative, low-cost fabrication.
- Program Arduino hardware to integrate servo and magnet control systems, and apply embedded systems logic to real-time hardware control.
- Contribute to a technical report to present findings at a future conference.

Earthquake Engineering Research Institute - Seismic Design Competition

Sep. 2025 - Present

Ground Motion Team

- Use the NHR3 NGA-Subduction Web Portal to identify and narrow down 11 representative ground motion records for GM#2 based on seismic hazard criteria.
- Design molds in AutoCAD and SolidWorks, and fabricate components via 3D printing and laser cutting to connect trusses during the structural build.

Ulta Beauty

Jun. 2024 – Present

Beauty Advisor

Lynnwood, WA

- Drive weekly financial goals by promoting Ulta Beauty Rewards and credit card sign-ups.

Meadowdale High School

Sep. 2021 - Jun. 2025

Key Club President | ASB Secretary | Peer Tutor Ambassador

Lynnwood, WA

- Secured event sponsorships through cold outreach, organized school-wide events, and managed budget allocation across activities.